

Segmentation Evaluation Report for Magnaporthe Growth Analysis

evaluation_report pipeline

April 7, 2026

1 Introduction

This report validates a hybrid segmentation system for Magnaporthe growth analysis on 22 matched annotation pairs out of 24 available predictions. The goal is to quantify how closely the final predicted segmentation contour matches the ground-truth annotation set.

2 Methods

2.1 Hybrid segmentation pipeline

The evaluated output is the final hybrid segmentation contour produced after classical segmentation, U-Net support, Gemma-guided decision logic when available, and SAM refinement. Ground-truth and prediction masks are compared directly as binary foreground maps.

2.2 Evaluation protocol

Each predicted annotation is matched to a ground-truth annotation by normalized filename stem. The polygons are rasterized into binary masks, aligned to a shared canvas when needed, and then evaluated pixel-wise. The dataset-level summary reports the mean, standard deviation, and bootstrap confidence interval of each metric.

3 Metric definitions

Let TP , FP , and FN denote true-positive, false-positive, and false-negative pixel counts.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN}$$

4 Results

4.1 Aggregate metrics

Metric	Mean	Std	CI lower	CI upper
precision	0.7272	0.2345	0.6254	0.8204
recall	0.9133	0.2160	0.8104	0.9789
f1_score	0.7988	0.2096	0.7020	0.8723
iou	0.7019	0.2298	0.6011	0.7911

4.2 Headline values

- Mean precision: 0.7272
- Mean recall: 0.9133
- Mean F1-score: 0.7988
- Mean IoU: 0.7019
- Missing prediction-to-GT matches: 2

4.3 Area agreement statistics

Statistic	Value
Mean predicted area (px)	467628.2727
Mean GT area (px)	405068.5455
Mean relative area error (%)	38.0582
Median relative area error (%)	30.3602
Shape alignment rate	0.7273

4.4 Figures

5 Discussion

The results summarize how well the final hybrid segmentation output aligns with manual ground truth on the available annotation set. Precision reflects over-segmentation control, recall reflects missed-colony risk, F1 balances both, and IoU gives a compact overlap-based score. Bootstrap confidence intervals provide a small-sample uncertainty estimate rather than a single optimistic point value.

5.1 Conclusions

Across 22 matched mask pairs out of 24 available predictions, the hybrid segmentation pipeline delivered moderate overlap with the manual labels. The mean IoU was 0.7019 with a bootstrap 95\% confidence interval of [0.6011, 0.7911], while the mean F1-score was 0.7988. Precision (0.7272) and recall (0.9133) show whether the system leans more toward under-segmentation or

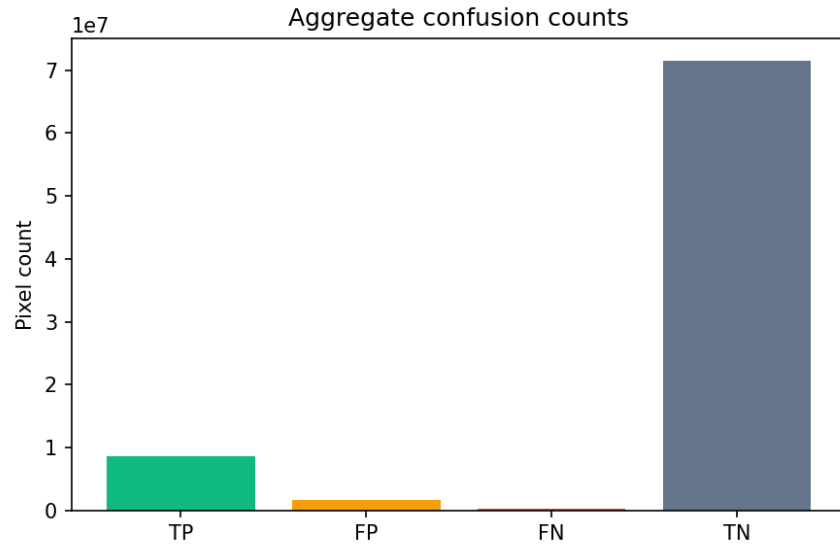


Figure 1: Aggregate confusion counts across the evaluation dataset.

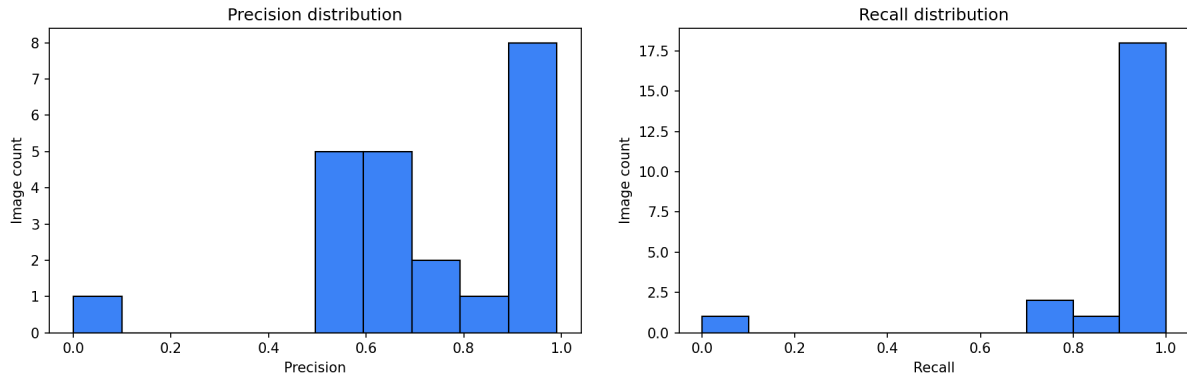


Figure 2: Precision and recall distributions.

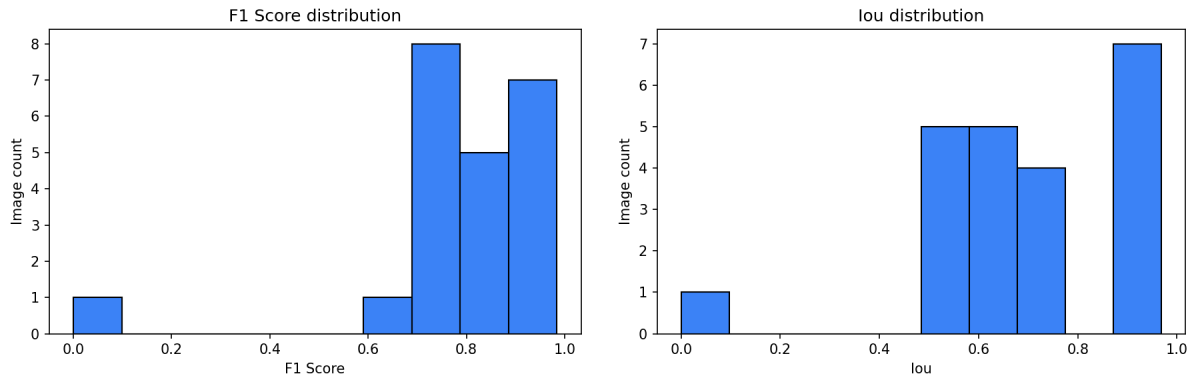


Figure 3: F1-score and IoU distributions.

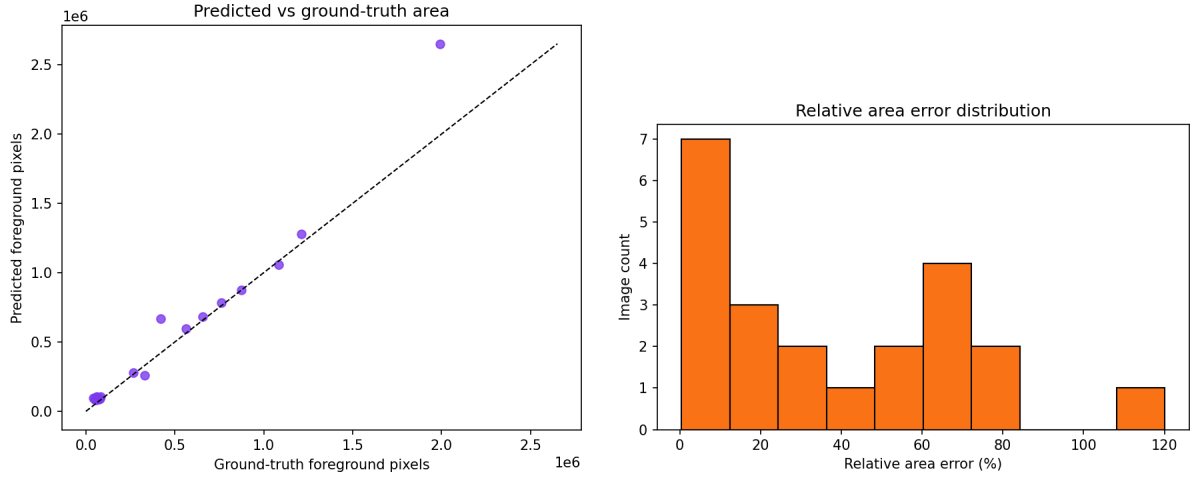


Figure 4: Annotation-derived area agreement and relative area error.

over-segmentation. The median relative foreground-area error was 30.36%, and 0.7273 of pairs required canvas-size alignment before pixel-wise comparison. These statistics suggest that the final performance estimate is driven by the full annotation set rather than a single optimistic example, but the unmatched files and the alignment rate should still be reviewed before treating the numbers as a final benchmark.